Topics

- Algorithm efficiency
- Running time
- Memory usage
- Input size
- Best case
- Average case
- Worst case
- Big-O notation
- `O(1)`
- `O(log n)`
- `O(n)`
- `O(n log n)`
- `O(n²)`

Laboratory

Students analyze a customer search algorithm.

```
def find_customer(customers, target):
    for customer in customers:
        if customer == target:
            return True
    return False
```

MIS Context

> What happens when a customer database grows from 100 records to 10 million records?